

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech CSE**

Semester: 6

Name of the Course: Devops on Cloud

Course Code: TCS 651

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	Evaluate the benefits of adopting DevOps in software projects when compared to the Agile framework.	
	OR	
(b)	Analyse how cloud computing enables businesses to scale resources dynamically and how DevOps practices contribute to faster application deployment with minimal downtime.	CO2
Q2	(10 Marks)	CO2
(a)	Explain how DevOps principles help improve collaboration between development and operations teams. How do these principles impact teamwork and project success?	
	OR	
(b)	Illustrate the DevOps Lifecycle and analyze how the implementation of DevOps principles influences the efficiency of cloud-based infrastructure.	CO4
Q3	(10 Marks)	CO2
(a)	Assess the impact of popular DevOps tools in areas such as automation, CI/CD, monitoring, alerting, version control, and scripting. How do these tools enhance the software development process?	
	OR	
(b)	Evaluate and distinguish between Continuous Integration (CI) and Continuous Deployment (CD). How do these practices enhance and support one another within the context of a DevOps?	CO3
Q4	(10 Marks)	CO1
(a)	Evaluate the role of version control in software development and assess how it contributes to the overall process. Justify the advantages of implementing version control systems like Git.	
	OR	
(b)	Compare and contrast Centralized Version Control System (CVCS) and Distributed Version Control System (DVCS). List the advantages and disadvantages of each approach.	CO2
Q5	(10 Marks)	CO3
(a)	Analyze the following Git concepts: (Any three) 1. Branch 2. Merge 3. Clone 4. Fork	
	OR	
(b)	Discuss how Jenkins aids in identifying and resolving integration issues early in the software development lifecycle. How does it improve development efficiency?	CO5

